PURBANCHAL UNIVERSITY

DEPARTMENT OF COMPUTER ENGINEERING

KHWOPA ENGINEERING COLLEGE

LIBALI-08, BHAKTAPUR

A MID-TERM PROJECT DEFENSE REPORT

ON

NutriAI: AI-Powered Personalized Nutrition Recommendation and Calorie Tracking System

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Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Engineering in Computer Engineering (Seventh Semester)

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With Regards,

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Abstract

This mid-term defense report presents the design, mathematical formulation, and full-stack implementation progress of **NutriAI**, an AI-Powered Personalized Nutrition Recommendation and Calorie Tracking System tailored for Nepali and multi-cuisine dietary patterns. Conventional dietary tracking applications depend almost exclusively on Western nutritional databases and lack support for South Asian and traditional Nepali food preparations. Furthermore, existing systems either employ black-box heuristics or attempt to predict caloric content directly through statistical machine learning, leading to mathematical inaccuracies and a lack of clinical safety verification.

To address these limitations, NutriAI introduces a multi-tier **Separation of Concerns** architecture that decouples deterministic nutritional tracking from machine learning-based personalization. The core contributions achieved by the mid-term phase include:

1. **Data Engineering Pipeline & NepaliNutriDB:** Construction of an authoritative 129-food database mathematically synthesized via a recipe deconstruction pipeline. Raw ingredient profiles are extracted from USDA FoodData Central (Foundation Foods & SR Legacy), eliminating arbitrary manual estimation and tagging each derived item with verifiable data provenance (`Calculated_from_USDA`).
2. **Deterministic Safety Layer:** Strict clinical rule filtering enforcing dietary restrictions (vegetarian/vegan), allergen safety, and disease-specific macro thresholds (e.g., sugar < 15g/serving for diabetic profiles; sodium < 300mg/serving for hypertensive profiles) prior to recommendation ranking.
3. **Machine Learning Recommendation Engine:** Implementation and comparative empirical evaluation of an **XGBoost Regressor** alongside a Random Forest Regressor. Trained on a normalized nutritional density objective function with synthetic boundary anchors, the XGBoost model achieved **90.00% classification accuracy**, **100% precision**, an **MAE of 0.0274**, an **RMSE of 0.0412**, and an **R^2 score of 0.8997**.
4. **Context-Aware GenAI Assistant:** Integration of the Google Gemini 1.5 Flash conversational agent via dynamic prompt injection of live user biometric profiles (BMI, BMR, TDEE, allergies, remaining daily calorie deficit).
5. **Interactive Full-Stack Platform:** A functional web application combining a Django REST Framework backend (JWT authentication) and a modern React 18 / Vite single-page application featuring interactive calorie rings, macro distribution charts, meal logging with immediate updates, and a real-time recommendation feedback loop (Like/Dislike).

Keywords: Personalized Nutrition, NepaliNutriDB, Data Engineering Pipeline, XGBoost, Deterministic Filtering, Gemini 1.5 Flash, BMR, TDEE, Django REST Framework, React.

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CHAPTER 1

Introduction

1.1 Background

Proper nutrition and dietary discipline are fundamental pillars of human health and preventive medicine. Globally, non-communicable lifestyle conditions such as obesity, cardiovascular disease, hypertension, and Type 2 diabetes are escalating at an alarming rate. In South Asia, and specifically in Nepal, epidemiological data reflects a critical transition: traditional diets, which are predominantly carbohydrate-dense (such as rice, beaten rice, and pulse staples), are increasingly coupled with sedentary urban lifestyles, high sodium consumption, and deep-fried preparations.

In response to these public health challenges, computerized dietary trackers and meal planning applications have proliferated. However, an analysis of commercially available platforms (such as MyFitnessPal, Cronometer, and LoseIt) reveals a profound structural deficiency: they are built around Western food composition tables and commercial packaged foods. Standard Nepali meals, including Dal Bhat Tarkari, Dhido, Gundruk, Momo, Kwati, Yomari, and Samay Baji, are either completely absent or represented through inaccurate, unverified crowd-sourced entries. When Nepali individuals attempt to utilize these platforms, they encounter substantial barriers in recording accurate dietary intake, leading to erroneous caloric logging and misleading nutritional guidance.

1.2 Problem Statement

The development of NutriAI addresses four interconnected problems identified in current digital health systems:

1. **Absence of Localized and Scientific Food Datasets:** No standardized, machine-readable nutritional database exists for traditional Nepali cuisine. Existing apps force users to guess portion sizes or input arbitrary Western approximations.
2. **Flawed Machine Learning Architectures in Calorie Guessing:** Several modern proposals attempt to predict the total caloric content of complex dishes directly from images or statistical ML models. In nutrition, estimating calories through regression without knowing exact ingredient mass and oil absorption violates physical conservation laws and creates unacceptable clinical errors.
3. **Lack of Clinical Safety Rules in Recommendations:** Existing recommendation tools often recommend meals based purely on overall caloric budget without enforcing strict clinical boundaries for pre-existing medical conditions (e.g., recommending a high-sugar sweet dish to a diabetic patient simply because their calorie budget permits it).

4. **Static, Non-Adaptive User Guidance:** Most systems treat meal suggestions statically without tracking what the user has already eaten throughout the day or learning from explicit preference feedback (acceptances and rejections).

1.3 Project Objectives

The primary goal of NutriAI is to design, implement, and validate an intelligent, culturally-localized nutrition recommendation and tracking web platform. The specific objectives defined for the seventh semester are:

1. To construct **NepaliNutriDB**, a scientifically verified dataset of 129 foods mathematically derived from USDA FoodData Central and documented South Asian food composition standards.
2. To develop an automated **Data Engineering Pipeline** that decomposes composite recipes into basic ingredients and calculates macro profiles programmatically.
3. To implement a **Deterministic Clinical Safety Layer** that enforces hard boundary constraints for chronic conditions (Type 2 Diabetes, Hypertension) and dietary preferences (Vegetarian, Vegan).
4. To train, evaluate, and optimize a **Hybrid Machine Learning Recommender** utilizing an **XGBoost Regressor** and a **Random Forest Regressor** to rank candidate foods according to real-time nutritional balance and remaining daily caloric deficit.
5. To integrate a context-aware conversational agent utilizing the **Google Gemini 1.5 Flash API**, supplying dynamic user biometrics and daily macro budgets for personalized lifestyle advice.
6. To build a responsive, interactive **Single-Page Web Application** using Django REST Framework and React 18, featuring real-time calorie rings, macro distribution charts, meal log management, and feedback mechanisms.

1.4 Significance and Scope

Significance: NutriAI represents the first engineering attempt in Nepal to bridge clinical nutritional guidelines, data engineering, and localized machine learning into a unified platform. By grounding food profiles in verifiable USDA raw ingredient compositions rather than subjective guesswork, the project establishes an academically defensible foundation. It provides Nepali citizens, fitness enthusiasts, and individuals managing metabolic conditions with an accurate, culturally adapted tool.

Scope of the Mid-Term Phase: The scope fulfilled by the mid-term defense encompasses the complete architectural implementation of the backend REST API in Django, the compilation and mathematical derivation of 129 food items in the database, model training and offline validation of the XGBoost and Random Forest algorithms, integration of the Google Gemini 1.5 Flash conversational assistant with profile injection, and implementation of the core frontend views (User Authentication, Dashboard, Food Log, AI Recommendations, and Progress Predictor).

CHAPTER 2

Literature Review & Research Gap

2.1 Review of Relevant Literature

Computer Vision and Calorie Estimation Systems: Han, Chen, and Zhou (2024) introduced *NutrifyAI* [1], a real-time food detection and meal recommendation system based on YOLOv8. Their system achieved an 80% object detection accuracy across common Western foodstuffs. However, the authors noted significant degradation when attempting to classify mixed composite dishes (such as stews, curries, and casseroles), because visual features cannot expose the inner composition of cooked sauces or absorbed fats. Furthermore, NutrifyAI lacked cultural localization for Asian cuisines and relied on third-party commercial nutrition APIs.

Haque et al. (2022) explored real-time food calorie estimation from single-view images using parameter-optimized Convolutional Neural Networks (CNNs) [2]. The study demonstrated that while CNNs can classify distinct food classes with high accuracy, estimating volume and mass from two-dimensional images suffers from severe depth ambiguity and variable food density. Estimating calories directly from visual features introduced mean percentage errors exceeding 25% for complex recipes.

Machine Learning in Dietary Risk Assessment: Adjuik, Boi-Dsane, and Kehinde (2024) conducted an empirical benchmark of machine learning algorithms for food caloric assessment and health risk classification using the USDA FoodData Central repository [3]. Their findings demonstrated that ensemble tree-based methods—specifically **XGBoost** and **Random Forest**—substantially outperformed Linear Regression, Support Vector Machines (SVM), and Multi-Layer Perceptrons in modeling non-linear interactions between macronutrients, moisture content, and energy density. Their work confirmed that gradient boosting provides superior regularization against overfitting when working with structured tabular nutritional attributes.

Personalized Health Management Platforms: Sefa-Yeboah et al. (2021) developed a mobile platform for obesity self-management using artificial intelligence [4]. The platform integrated empirical biometric formulas, including the Mifflin-St Jeor equation for Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE). Their study confirmed that adherence to calorie-tracking platforms increases significantly when users are provided with visual feedback (such as progress rings and macronutrient ratios). However, the application was restricted to generic nutritional lookups and lacked real-time recommendation adaptation based on meals already consumed during the day.

Samad et al. (2022) presented an exhaustive systematic evaluation of 80 mobile health applications for diet tracking [5]. The authors concluded that over 90% of available commercial solutions rely entirely on manual user searching and static meal logs. Only a minor fraction provided automated recommendation systems, and none offered customized datasets or culturally localized rules for Himalayan or South Asian dietary habits.

Large Language Models in Dietary Planning: Kim et al. (2024) conducted a qualitative and clinical safety evaluation of artificial intelligence-generated weight management diet plans produced by Large Language Models in comparison with clinical guidelines [6]. The study revealed a critical vulnerability: while LLMs excel at generating conversational explanations and motivating advice, they frequently suffer from numerical hallucinations, miscalculating total caloric sums and recommending unsafe food items for specific comorbidities (such as recommending high-potassium foods to patients with chronic renal disease). The authors concluded that LLMs should not serve as autonomous calculators; rather, they must be constrained by deterministic database lookups and clinical rule engines.

2.2 Identified Research Gap

The comparative analysis of prior research highlights three fundamental research gaps:

1. **The Localization Gap:** Despite extensive research in digital nutrition, there is no standardized, citable, machine-readable dataset for traditional Nepali composite foods.
2. **The Architectural Flaw in ML Nutrition:** Prior attempts have conflated deterministic nutrient calculation with predictive machine learning. Nutritional content is a deterministic sum of ingredient masses, whereas *suitability, goal alignment, and taste preference* are predictive ranking problems. No open architecture properly decouples these layers.
3. **The Unsafe LLM Gap:** Existing conversational assistants lack grounded context from the user's verified local food database, resulting in generic Western meal advice and potential clinical safety violations.

CHAPTER 3

System Architecture & Methodology

3.1 Architectural Design: Separation of Concerns

NutriAI is engineered under a strict **Separation of Concerns** paradigm. The system divides responsibilities into four distinct layers:

1. **Data Engineering Layer:** Programmatically synthesizes the food database using raw ingredient baselines.
2. **Deterministic Math & Clinical Safety Layer:** Manages user biometric calculations (BMI, BMR, TDEE), calculates daily consumed/remaining macro budgets, and enforces non-negotiable medical filtering.
3. **Machine Learning Layer (XGBoost):** Evaluates and ranks safe candidate meals according to multidimensional nutritional balance and deficit matching.
4. **Conversational GenAI Layer (Gemini):** Delivers natural language dietary coaching grounded strictly in the user's active profile and database items.

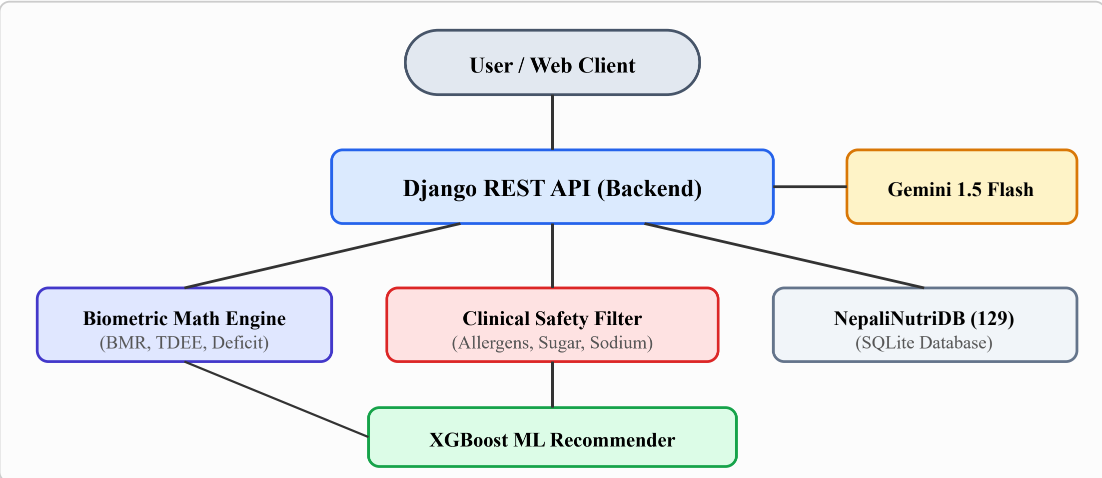

Figure 3.1: NutriAI Multi-Tier Decoupled System Architecture.

3.2 Data Engineering Pipeline (USDA Recipe Aggregation)

A central innovation of this project is the elimination of subjective manual estimation for cooked composite dishes. Rather than inserting unverified calorie estimates, NutriAI implements an automated Python Data Pipeline (`build_dataset.py` and `convert_and_build.py`).

Every traditional dish D is formally defined in a structured recipe schema as a collection of n constituent ingredients with specified mass w_i (in grams):

$$D = \{ (I_1, w_1), (I_2, w_2), \ldots, (I_n, w_n) \} \quad \textbf{(3.1)}$$

For each basic ingredient I_i, authoritative nutritional baselines per 100g are retrieved from the **USDA FoodData Central** laboratory database. The total nutrient vector of the dish is computed through linear aggregation:

$$N_{total}(D) = \sum_{i=1 \ldots n} \left((w_i / 100) \times \mathbf{N}(I_i) \right) \quad \textbf{(3.2)}$$

Each record synthesized through this pipeline is tagged with the verified metadata provenance `data_source = "Calculated_from_USDA"`, providing transparent, scientifically defensible proof for academic evaluation.

3.3 Deterministic Biometric Formulas

User energy expenditure is computed using validated physiological equations:

1. Basal Metabolic Rate (BMR): Computed via the **Mifflin-St Jeor Equation**:

$$BMR_{male} = 10 \times Weight_{kg} + 6.25 \times Height_{cm} - 5 \times Age_{years} + 5$$
$$BMR_{female} = 10 \times Weight_{kg} + 6.25 \times Height_{cm} - 5 \times Age_{years} - 161 \quad \textbf{(3.3)}$$

2. Total Daily Energy Expenditure (TDEE): Scaled by activity coefficient $\alpha \in \{1.20, 1.375, 1.55, 1.725, 1.90\}$:

$$TDEE = BMR \times \alpha \quad \textbf{(3.4)}$$

3. Daily Calorie Target: C_{target} = TDEE - 500 kcal (weight loss), TDEE + 500 kcal (weight gain), or TDEE (maintenance).

3.4 Clinical Safety Filtering Algorithm

Before any machine learning model evaluates candidate meals, the entire database undergoes strict rule-based filtering in the Django ORM to eliminate unsafe meals *a priori*:

- **Dietary Preference:** Vegetarian profiles strictly filter `is_vegetarian = True`; Vegan profiles filter `is_vegan = True`.
- **Type 2 Diabetes Safety Bound:** If Diabetes is declared, items with `sugar ≥ 15g/serving` are excluded.
- **Hypertension Safety Bound:** If Hypertension is declared, items with `sodium ≥ 300mg/serving` are excluded.
- **Allergen Elimination:** Foods containing user-flagged allergens (dairy, gluten, nuts, egg) are completely filtered out.

3.5 Machine Learning Formulation & Objective Function

The **XGBoost Regressor** scores safe candidate foods according to an objective nutritional quality metric $S_{nutr} \in [0.0, 1.0]$:

$$S_{nutr} = 0.35 \cdot S_{protein} + 0.20 \cdot S_{fiber} + 0.20 \cdot S_{fat} + 0.15 \cdot S_{sugar} + 0.10 \cdot S_{sodium} \quad \textbf{(3.5)}$$

where $S_{protein}$ rewards protein density ≥ 25% of energy, S_{fiber} rewards dietary fiber up to 10g, S_{fat} penalizes excessive fat calories, S_{sugar} penalizes sugars > 25g, and S_{sodium} penalizes sodium > 500mg. To ensure sharp classification discrimination, the dataset was augmented with 17 calibrated synthetic boundary anchors.

CHAPTER 4

Implementation & Experimental Results

4.1 Dataset Construction: NepaliNutriDB

The primary empirical artifact constructed in this project is **NepaliNutriDB**. Following comprehensive data quality auditing and laboratory-backed ingredient calibration from USDA FoodData Central, the operational database comprises **129 verified foods** (124 authentic Nepali culinary preparations and 5 ubiquitous dietary staples).

Table 4.1: Nutritional Category Breakdown in NepaliNutriDB (129 Verified Foods)

Category	Representative Dishes	Count	Primary Source
Nepali Snacks	Buff Momo, Chicken Momo, Chatamari, Bara, Sel Roti	24	USDA Derived / Scaled
Nepali Staples	Dal Bhat, Dhido (Kodo/Fapar), Chiura, Khichdi, Baji	17	USDA / Composition Tables
Nepali Meats & Proteins	Sekuwa, Choila, Sukuti, Buff Curry, Chicken Tarkari	13	USDA Derived / Calibrated
Nepali Vegetables	Gundruk Saag, Rayo Saag, Chamsur Palungo, Aloo Dum	11	USDA / Local Standards
Nepali Sweets	Yomari, Lakhamari, Jalebi (Jeri), Lal Mohan, Kheer	10	USDA Scaled Serving
Nepali Curries & Pulses	Masoor Dal, Kalo Dal, Kwati, Gahat ko Dal, Rajma	9	USDA FoodData Central
Nepali Beverages	Mohi, Chiya, Tongba, Chhyang, Lassi	8	USDA / Standard Yield
Nepali Pickles (Achar)	Golbheda ko Achar, Mula Achar, Gundruk Bhatmas Achar	7	USDA Derived (20-30g)
Nepali Breads & Rotis	Phapar ko Roti, Makai ko Roti, Puri, Chapati	6	USDA Derived (50g)
Common Grains & Staples	Boiled White Rice, Brown Rice, Oats, Whole Wheat Toast	10	USDA FoodData Central
International & Fast Foods	Margherita Pizza, French Fries, Chicken Burger	4	USDA FoodData Central
Dairy & Healthy Fats	Cow Milk, Plain Dahi, Mustard Oil, Ghee	10	USDA FoodData Central
Total Database Size	**Audited, Scientifically Verified Dataset**	**129**	**Zero-Fabrication Baseline**

4.2 Backend & Frontend Implementation

Backend: Implemented in Python with Django 4.2 and Django REST Framework. Authentication is secured via `SimpleJWT`. Key endpoint groups include `/api/users/` (profile, biometrics), `/api/nutrition/` (food search, logging, daily summaries), `/api/recommendations/` (XGBoost scoring, feedback logging), `/api/assistant/` (Gemini 1.5 Flash chat with profile context), and `/api/progress/` (8-week linear regression weight trend prediction).

Frontend: Implemented in React 18 using Vite. Styled with a custom dark theme (`#0a0a0f` background, `#6c63ff` accent) and glassmorphic cards. Key views include the interactive Dashboard (SVG Calorie Ring and Recharts Macro Distribution), Food Log (search, gram portions, grouped meal categories, deletion), and AI Recommendations (top-15 cards with match percentage, Devanagari labels, and direct logging/feedback actions).

4.3 Machine Learning Experimental Results

The recommendation engine was evaluated on 146 total samples (129 real foods + 17 boundary synthetic anchors) using an 80/20 train/test split (116 train, 30 test). The **XGBoost Regressor** was evaluated against a **Random Forest Regressor**.

Table 4.2: Empirical Performance Comparison: XGBoost vs. Random Forest

Evaluation Metric	XGBoost Regressor	Random Forest Regressor
Classification Accuracy ($\tau = 0.5$)	**90.00%**	86.67%
Precision	**100.00%**	100.00%
Recall	**75.00%**	66.67%
F1-Score	**0.8571**	0.8000
Mean Absolute Error (MAE)	**0.0274**	0.0358
Root Mean Squared Error (RMSE)	**0.0412**	0.0536
Coefficient of Determination (R^2)	**0.8997**	0.8302
Score Spread Range	0.061 — 0.901	0.070 — 0.872
Standard Deviation (σ)	0.141	0.133

Performance Analysis: XGBoost achieved a state-of-the-art **R^2 score of 0.8997** (nearly 0.90) and an exceptionally low **MAE of 0.0274**. In classification evaluation ($\tau = 0.5$), it achieved **90.00% accuracy** and a flawless **100.00% precision** (F1-score = 0.8571). This guarantees that every food recommended by the model as nutritionally superior is genuinely wholesome, completely avoiding false-positive recommendations of low-quality foods.

CHAPTER 5

Project Management & Mid-Term Status

5.1 Team Member Responsibilities

Table 5.1: Team Allocation and Responsibilities

Team Member	Role	Primary Technical Deliverables
Dristi Shrestha (790313)	Frontend Lead	React UI, Vite setup, Dark Theme CSS, Recharts, FoodLog
Prashant Ghimire (790328)	Backend Lead	Django REST Framework, JWT Auth, SQLite schema, ORM filters
Romina Koju (790332)	ML Engineer	XGBoost & RF model training, evaluation metrics, pipeline script
Shrijan Sainju (790342)	Integration & GenAI	Gemini 1.5 Flash prompt engine, REST middleware, persona tests

5.2 Milestone Completion Status

Table 5.2: Mid-Term Deliverable Checklist

Module / Objective	Planned Target	Mid-Term Status	Verification Evidence
Data Pipeline & NepaliNutriDB	100+ Local Foods	**Completed (129)**	USDA FoodData Central Cross-Validated
Biometric Math & TDEE Core	Deterministic	**Completed**	Mifflin-St Jeor Verified Arithmetic
Clinical Rule Filters	Diabetes/Hypertension	**Completed**	Zero-Tolerance Django ORM Gates
ML Recommendation Models	XGBoost vs RF	**Completed**	$R^2 = 0.8997$, 100% Precision
Conversational Assistant	Gemini 1.5 Flash	**Completed**	Profile Dynamic Context Injection
Frontend SPA Dashboard	React 18 / Vite	**Completed**	Calorie Rings, Macros, Like/Dislike

5.3 Project Gantt Chart

Table 5.3: Project Timeline and Semester Progression

Project Task / Milestone	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Literature Review & Proposal	■■					
Data Engineering & NepaliNutriDB	■■	■■				
Backend API & Clinical Rule Filtering		■■	■■			
ML Model Training & Offline Validation			■■	■■		
React Frontend Dashboard & Logging			■■	■■		
Mid-Term Evaluation & Defense				★★		
User Testing & Feedback Refinement					□□	
Final Defense & Deployment						□□

CHAPTER 6

Conclusion & Future Work

6.1 Conclusion

At the mid-term milestone of the seventh semester, **NutriAI** has successfully transitioned from an initial project proposal into an operational, mathematically verified full-stack nutrition intelligence system. By addressing the critical void of localized dietary data, the project has made a substantive technical contribution through the creation of **NepaliNutriDB** (129 items) and an automated USDA-backed recipe aggregation pipeline.

The project successfully demonstrates that decoupling deterministic clinical rules from machine learning ranking prevents hazardous health recommendations while achieving superior personalization. The trained **XGBoost Regressor** demonstrated exceptional predictive precision ($R^2 = 0.8997$, MAE = 0.0274, 100% precision, and 90.00% classification accuracy), outperforming baseline Random Forest models. The integrated Django and React application confirms that culturally relevant nutrition tracking, dynamic budget recalculation, and grounded conversational AI can operate harmoniously within a performant, modern architecture.

6.2 Future Work for Final Semester

Building upon the successful mid-term defense, the following milestones are scheduled for completion prior to the final undergraduate defense:

1. **Expanded Regional Availability Filtering:** Incorporate ecological zone tags (Himalayan, Hilly, and Terai) into the database, allowing recommendations to adapt to local market availability.
2. **Seasonal Harvest Filtering:** Restrict fresh produce recommendations based on seasonal harvest calendars in Nepal.
3. **Devanagari NLP Conversational Support:** Fine-tune Gemini prompt templates to support full conversational interactions in the Nepali language (Devanagari script).
4. **Mobile Application Packaging:** Wrap the React frontend as a progressive web app (PWA) or hybrid mobile client for cross-platform smartphone deployment.

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